3.1–3.2 · Optimization I: descent, step size, and the damped Gauss-Newton step for inverse kinematics

Sessions 3.1–3.2 use a scalar cost to turn inverse kinematics into a numerical optimization problem. We begin with gradient descent and its step size, then use the linearized residual to derive a damped Gauss–Newton step.

We build on forward kinematics and the Jacobian from sessions 2.3–2.4. At the current joint positions, the Jacobian predicts the effect of a small joint change. This reading derives how to choose that change and decide how far to apply it. The generic decision variable is $\mathbf{z}$; in inverse kinematics it is the joint configuration $\mathbf{q}$. The cost f is a scalar, and $\boldsymbol{\delta}$ is a proposed model step. The letter d counts decision variables, whereas $\mathbf{d}$ denotes a search direction.

0. What you will be able to do afterwards

- Write inverse kinematics as an unconstrained minimization, write its gradient $\mathbf{J}^\top(\varphi - \mathbf{y}^*)$ and check it by finite differences, take gradient-descent steps on it, and choose the step size by Armijo backtracking rather than by trial and error.
- Predict from the Hessian's eigenvalues how a fixed step size behaves (contract, oscillate, diverge) and why the elongated bowl defeats any single step size.
- Derive Newton's step from the quadratic model; derive the Gauss-Newton step as the minimizer of the least-squares problem that the linearized residual defines, and say which term of the exact Hessian it drops; derive the damped step $\mathbf{q} \leftarrow \mathbf{q} + \mathbf{J}^\sharp(\mathbf{y}^* - \varphi(\mathbf{q}))$, $\mathbf{J}^\sharp = (\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})^{-1}\mathbf{J}^\top\mathbf{K}$, from the inverse-kinematics cost, line by line; state what each of $\mathbf{K}$, $\mathbf{D}$, and ρ does.
- Run that step on the two-link arm by hand for one iteration and by code to convergence, recognize quadratic versus linear convergence in a residual table, end every run with one of the four outcomes (solved, stationary, stalled, iteration limit) and both norms, and explain what happens at an unreachable target with a line search and with damping.

1. Motivation and analogy

The first local IK step leaves a task error even when the target is reachable. At the worked configuration from Deck 03, the current tip and desired position are

$$\varphi(\mathbf{q}^{(0)}) = \begin{bmatrix} 0.4241 \\ 0.4898 \end{bmatrix}\ \mathrm{m}, \qquad \mathbf{y}^* = \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix}\ \mathrm{m}.$$

Solving the square local model proposes a joint change of $(-52.7^\circ, 74.1^\circ)$. Applying it leaves the tip about 0.17 m from the target. The change is too large for the local prediction to remain accurate. A Panda position Jacobian also has three rows and seven columns, so an ordinary inverse is unavailable.

Instead, measure the position error with a cost:

$$f(\mathbf{q}) = \tfrac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2.$$

Reducing this cost provides a common numerical formulation for both arms. A local solver seeks a lower cost from its current guess; it does not certify a global minimum.

A hillside gives a useful picture of this search. The current joint configuration is a point on the hillside, and the cost is its height. The gradient gives the local uphill direction. Gradient descent moves the other way. Armijo checks the actual fall in cost before accepting a step. Newton uses local curvature to fit a quadratic model; Gauss–Newton builds that model from a linearized residual.

The search takes place in joint space. A straight change of joint angles generally moves the tip along a curved path in physical space. The left panel of `fig/ik_iterates.png` shows the two-link cost over joint angles, with two target-reaching minima and a saddle between them.

The hillside picture describes only local information. A small gradient can mark a minimum, a saddle or a maximum. A reachable IK target can therefore leave a solver at a stationary configuration with task error remaining. Section 5(p) demonstrates this case at the stretched arm. Damping controls the proposed step, but does not provide a reachability certificate.

2. Prerequisite ramp

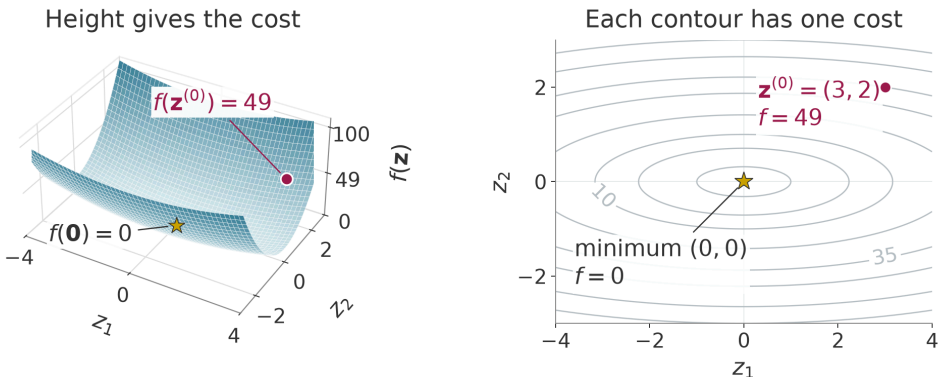
A cost surface before its derivatives. Consider the two-variable cost

$$f(\mathbf{z}) = z_1^2 + 10z_2^2, \qquad \mathbf{z}^{(0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

The coordinates z_1 and z_2 are the variables we can change. The height above each pair is its cost. At the starting point,

$$f(3, 2) = 3^2 + 10(2^2) = 49.$$

Both squared terms are nonnegative, so the minimum is at $(0, 0)$ with cost zero. A contour joins points with equal cost. The surface and contour map below show the same function and the same starting point. The coefficient 10 makes the valley narrower along z_2. These are decision-space coordinates; when we later set $\mathbf{z} = \mathbf{q}$, they are joint angles rather than tip coordinates.



The cost surface and its equal-cost contours, with the current point at $(3, 2)$ and the minimum at $(0, 0)$.

Slope gives the gradient. To find how cost changes when only z_1 moves, hold $z_2 = 2$. The resulting slice is

$$f(z_1, 2) = z_1^2 + 40, \qquad \left.\frac{\partial f}{\partial z_1}\right|_{(3,2)} = 2(3) = 6.$$

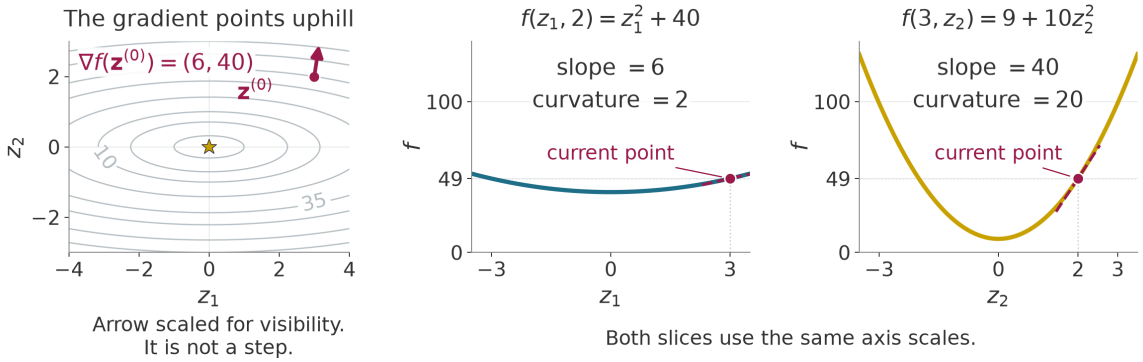
Holding $z_1 = 3$ instead gives the other slice:

$$f(3, z_2) = 9 + 10z_2^2, \qquad \left.\frac{\partial f}{\partial z_2}\right|_{(3,2)} = 20(2) = 40.$$

Stack the two partial derivatives to obtain the gradient:

$$\nabla f(\mathbf{z}) = \begin{bmatrix} \partial f/\partial z_1 \\ \partial f/\partial z_2 \end{bmatrix} = \begin{bmatrix} 2z_1 \\ 20z_2 \end{bmatrix}, \qquad \nabla f(3, 2) = \begin{bmatrix} 6 \\ 40 \end{bmatrix}.$$

For example, a small positive change ε in z_1 alone changes the cost by approximately 6ε at this point; the same change in z_2 changes it by approximately 40ε. The gradient points in the direction of fastest local increase for a unit change in the Euclidean norm of these coordinates. It is perpendicular to the equal-cost contour. The arrow below shows that direction in the decision plane, scaled for drawing. It is not a chosen step size.



The uphill gradient and two coordinate slices through the same current point. Each slice has cost 49 at the marked point; its tangent gives the current slope.

Changes of slope give the Hessian. The current slopes are 6 and 40, but their rates of change are 2 and 20. Differentiate the slopes once more:

$$\mathbf{H}(\mathbf{z}) = \nabla^2 f(\mathbf{z}) = \begin{bmatrix} \partial^2 f/\partial z_1^2 & \partial^2 f/(\partial z_1 \partial z_2) \\ \partial^2 f/(\partial z_2 \partial z_1) & \partial^2 f/\partial z_2^2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 20 \end{bmatrix}.$$

Entry (i, j) describes how slope i changes when coordinate j changes. The mixed entries are zero here because each squared term depends on just one coordinate. The gradient changes with $\mathbf{z}$, while this quadratic's Hessian is constant. In an inverse-kinematics cost, both generally depend on the current joint configuration. The next expansion combines cost, slope and curvature into a model for a small proposed change.

Taylor expansion to second order. For a smooth $f : \mathbb{R}^d \to \mathbb{R}$ and a small step $\boldsymbol{\delta}$,

$$f(\mathbf{z} + \boldsymbol{\delta}) = f(\mathbf{z}) + \nabla f(\mathbf{z})^\top \boldsymbol{\delta} + \tfrac{1}{2} \boldsymbol{\delta}^\top \mathbf{H}(\mathbf{z}) \boldsymbol{\delta} + O(\|\boldsymbol{\delta}\|^3), \tag{1}$$

where $\nabla f(\mathbf{z}) \in \mathbb{R}^d$ is the gradient (the vector of partial derivatives, pointing uphill) and $\mathbf{H}(\mathbf{z}) = \nabla^2 f(\mathbf{z}) \in \mathbb{R}^{d \times d}$ is the Hessian (the symmetric matrix of second partials). Deck 03 linearized a vector map, $\varphi(\mathbf{q} + \Delta\mathbf{q}) \approx \varphi(\mathbf{q}) + \mathbf{J}\Delta\mathbf{q}$; (1) is the same idea for a scalar, one term further. Truncating after the quadratic term gives the model that Newton minimizes (§4.4). Gradient descent uses only the linear term, and only for the direction: a linear model has no minimum, and the step $-\alpha \nabla f$ is the minimizer of $f + \nabla f^\top \boldsymbol{\delta} + \|\boldsymbol{\delta}\|^2/(2\alpha)$, the linear model plus a penalty on the step's length (§4.6 uses the same construction for the damped step).

Positive definiteness and eigenvalues. A symmetric $\mathbf{H}$ is positive definite when $\boldsymbol{\delta}^\top \mathbf{H} \boldsymbol{\delta} > 0$ for every $\boldsymbol{\delta} \neq 0$, equivalently when all its eigenvalues λ_i are positive; then the quadratic model has a unique minimum. The eigenvectors of a symmetric $\mathbf{H}$ are its curvature directions, which can always be chosen orthogonal to one another, and the eigenvalues are the curvatures along them; for the diagonal bowl of §4.1 they are the coordinate axes with curvatures 2 and 20. The **condition number** $\operatorname{cond}(\mathbf{H}) = \lambda_{\max}/\lambda_{\min}$ measures how stretched the bowl is: 1 for a round bowl, large for a narrow valley.

Sums of squares. Many robotics costs have the form $f(\mathbf{z}) = \tfrac{1}{2}\|\mathbf{r}(\mathbf{z})\|^2 = \tfrac{1}{2}\sum_i r_i(\mathbf{z})^2$ with a residual vector $\mathbf{r} : \mathbb{R}^d \to \mathbb{R}^m$: distances to targets, penetration depths, accelerations along a trajectory. Their gradient and Hessian have a special structure (§4.5) that Gauss-Newton exploits.

3. Formal setup

3.1 The unconstrained problem and descent methods

$$\min_{\mathbf{z}\in\mathbb{R}^d} f(\mathbf{z}). \tag{2}$$

The cost is smooth, and we can evaluate its value, gradient and sometimes Hessian. Here d is the number of decision variables: $d = n$ when $\mathbf{z} = \mathbf{q}$ is one configuration, and $d = nT$ for a trajectory of T free knots (Deck 05). The joint count keeps the letter n.

Each step makes two choices. The search direction $\mathbf{d}$ specifies a proposed change of the variables. The positive step size α scales that change, so the actual displacement is

$$\alpha\mathbf{d}.$$

For example, at the bowl's starting point, gradient descent chooses

$$\mathbf{d} = -\nabla f(3, 2) = \begin{bmatrix} -6 \\ -40 \end{bmatrix}.$$

With $\alpha = 0.01$, the actual change is

$$\alpha\mathbf{d} = \begin{bmatrix} -0.06 \\ -0.40 \end{bmatrix}.$$

The vector $\mathbf{d}$ need not have unit length. We want a direction that lowers the cost for a sufficiently small positive step. This is called a **descent direction**, and its formal test at the current point is

$$\nabla f(\mathbf{z})^\top \mathbf{d} < 0.$$

A large step can still overshoot, so choosing a descent direction does not settle the step size. Every method below chooses a direction and a step size, then moves to

$$\mathbf{z} + \alpha\mathbf{d}.$$

- **Gradient descent:** $\mathbf{d} = -\nabla f(\mathbf{z})$.
- **Newton:** $\mathbf{d} = -\mathbf{H}(\mathbf{z})^{-1}\nabla f(\mathbf{z})$, a descent direction when $\mathbf{H}$ is positive definite.
- **Gauss-Newton and damped Gauss-Newton (fixed ρ):** Newton with the Hessian of a sum of squares replaced by $\mathbf{J}_r^\top\mathbf{J}_r$ (+ $\rho\mathbf{I}$); Levenberg–Marquardt proper adapts ρ (§4.6).

Three useful properties are **monotone** decrease, **convergence** to stationarity, and **robustness** to rescaling of the cost or variables. A fixed gradient-descent step does not guarantee these properties for every smooth cost. With Armijo, gradient descent guarantees monotone decrease and a gradient tending to zero when the cost is

bounded below and its gradient is Lipschitz. This theoretical statement uses the direction $-\nabla f$ and backtracking without an implementation floor. Section 4.3 gives the decrease bound. More general directions need additional conditions on their angle and size; a descent direction alone is insufficient.

A gradient tending to zero does not by itself establish an attained limit point. For example, the scalar cost

$$f(z) = \log(1 + e^{-z})$$

satisfies the smoothness and lower-bound assumptions, but its descent iterates can escape to positive infinity. A bounded sublevel set prevents that escape. The planar arm's hinge angles taken modulo 2π form a compact torus. Its unwrapped angles in $\mathbb{R}^n$ have unbounded sublevel sets because the cost repeats every revolution.

3.2 The step size: Armijo's condition

Given $\mathbf{z}$, a descent direction $\mathbf{d}$, and a trial step α, accept the step if

$$f(\mathbf{z} + \alpha\mathbf{d}) \le f(\mathbf{z}) + c\alpha\nabla f(\mathbf{z})^\top \mathbf{d}, \qquad 0 < c < 1. \tag{3}$$

We use $c = 0.01$. The actual decrease must be at least a fraction c of the drop the linear model predicts. **Backtracking**: start from $\alpha = 1$ (or the last accepted step grown by 1.2), halve until (3) holds, then move.

3.3 Least squares, Gauss-Newton, damped Gauss-Newton

$$f(\mathbf{z}) = \tfrac{1}{2}\|\mathbf{r}(\mathbf{z})\|^2, \mathbf{r} : \mathbb{R}^d \to \mathbb{R}^m, \mathbf{J}_r(\mathbf{z}) = \partial\mathbf{r}/\partial\mathbf{z} \in \mathbb{R}^{m\times d}. \tag{4}$$

The **Gauss-Newton step** minimizes the least-squares model $\tfrac{1}{2}\|\mathbf{r} + \mathbf{J}_r\boldsymbol{\delta}\|^2$ of the linearized residual, which gives the normal equations $(\mathbf{J}_r^\top\mathbf{J}_r)\boldsymbol{\delta} = -\mathbf{J}_r^\top\mathbf{r}$ (§4.5); the **damped Gauss-Newton step** adds $\tfrac{1}{2}\rho\|\boldsymbol{\delta}\|^2$ to the model and solves $(\mathbf{J}_r^\top\mathbf{J}_r + \rho\mathbf{I})\boldsymbol{\delta} = -\mathbf{J}_r^\top\mathbf{r}$ with a fixed damping $\rho > 0$; Levenberg–Marquardt methods change ρ from step to step, and the simple rule of §4.6 is one of them.

3.4 Inverse kinematics as a least-squares problem

$$\min_{\mathbf{q}} f(\mathbf{q}) = \tfrac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|_{\mathbf{K}}^2, \|\mathbf{v}\|_{\mathbf{M}}^2 := \mathbf{v}^\top\mathbf{M}\mathbf{v}, \tag{5}$$

with $\mathbf{K}$ positive definite weighting the task error (the essentials' weight w of §3). $\mathbf{K}$ is a choice you make about how much each task direction counts, and the units of the task coordinates do not determine it. Each step of the method solves the linearization of (5) at the current iterate $\mathbf{q}^{(i)}$, with a penalty on the step $\boldsymbol{\delta} = \mathbf{q} - \mathbf{q}^{(i)}$:

$$\min_{\boldsymbol{\delta}} \tfrac{1}{2}\|\varphi(\mathbf{q}^{(i)}) + \mathbf{J}\boldsymbol{\delta} - \mathbf{y}^*\|_{\mathbf{K}}^2 + \tfrac{1}{2}\|\boldsymbol{\delta}\|_{\mathbf{D}}^2. \tag{5'}$$

The matrix $\mathbf{D}$ is positive semidefinite. The first term asks the tip to reach $\mathbf{y}^*$; the second term trades a little of that against a shorter step, and with $\mathbf{D}$ positive definite it keeps each step well posed when $\mathbf{J}$ is rank deficient or $n > m$. The iterate $\mathbf{q}^{(i)}$ is reset at every step, so $\mathbf{D}$ penalizes the step away from the current iterate: the essentials' term $\|\mathbf{q} - \mathbf{q}_0\|^2$ with a **fixed** $\mathbf{q}_0$ (§5 there) trades target accuracy against a preferred posture and changes the answer, and it is a different problem (Deck 05 uses it as a penalty relaxation). In this course $\mathbf{K} = \mathbf{I}$ and $\mathbf{D} = \rho\mathbf{I}$ unless stated; KOMO later weights cost features in the same way: its `scale` argument is squared in the cost.

Optional detail on vanishing damping. Write the step weight as

$$\mathbf{D} = \eta\mathbf{D}_0, \qquad \mathbf{D}_0 \succ 0.$$

As η tends to zero, the step approaches the smallest $\mathbf{D}_0$-weighted step among minimizers of the task-weighted least-squares model. It achieves zero modeled error only when the linearized request is attainable.

Units on the arm. f is in m^2, $\nabla f = \mathbf{J}^\top(\varphi - \mathbf{y}^*)$ in $\mathrm{m}^2/\mathrm{rad}$, the Hessian and ρ (which is added to $\mathbf{J}^\top\mathbf{J}$) in $\mathrm{m}^2/\mathrm{rad}^2$, and a gradient step size α in $\mathrm{rad}^2/\mathrm{m}^2$. That is why a fixed cap on α, or a fixed α, suits no problem in general (§4.3), and why "millimeters next to meters" (§7) matters: rescaling a coordinate rescales the curvature in that direction.

Position only. The worked IK examples use a position target, so $\mathbf{J}$ is 2×2 on the two-link arm and 3×7 on the Panda. A full-pose residual needs its own derivative, that of the chosen orientation error; the geometric angular Jacobian block is only that derivative's local approximation near agreement, and $\mathbf{K}$ then weighs radians against meters. Libraries and KOMO build it; nothing in this block derives it.

4. Derivation

4.1 Gradient descent on a quadratic: different coordinate scales

The same step size multiplies both gradient entries, but those entries change at different rates. We use the bowl to see what this means for each coordinate:

$$f(\mathbf{z}) = z_1^2 + 10z_2^2, \qquad \nabla f(\mathbf{z}) = \begin{bmatrix} 2z_1 \\ 20z_2 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 2 & 0 \\ 0 & 20 \end{bmatrix}.$$

This is the "Elongated Quadratic" in the web app. At iteration i, subtract the scaled gradient from the current point. Written one coordinate at a time, the update is

$$z_1^{(i+1)} = z_1^{(i)} - \alpha\big(2z_1^{(i)}\big),$$
$$z_2^{(i+1)} = z_2^{(i)} - \alpha\big(20z_2^{(i)}\big).$$

The superscripts distinguish the current point from the next point. Both right-hand sides use the current point. For this quadratic, the gradient equals the Hessian times the current point, so the same update has the matrix form

$$\mathbf{z}^{(i+1)} = \mathbf{z}^{(i)} - \alpha\nabla f(\mathbf{z}^{(i)}) = (\mathbf{I} - \alpha\mathbf{H})\mathbf{z}^{(i)}, \tag{6}$$

Factoring the current coordinate out of each subtraction shows what fraction of its value remains after one step:

$$z_1 \leftarrow (1 - 2\alpha)z_1, z_2 \leftarrow (1 - 20\alpha)z_2. \tag{7}$$

A factor of zero takes a coordinate to zero immediately. A factor with magnitude less than one makes repeated values approach zero. A negative factor crosses zero at each step; the magnitude determines whether those crossings get smaller or larger. For example, a factor of -0.6 gives a decaying oscillation, while a factor of -1 preserves its amplitude.

The two coordinate curvatures are the Hessian's eigenvalues, 2 and 20. For a positive curvature λ_i, the repeated factor is

$$1 - \alpha\lambda_i.$$

Convergence requires

$$|1 - \alpha\lambda_i| < 1 \quad \Longleftrightarrow \quad 0 < \alpha < \frac{2}{\lambda_i}.$$

Setting this factor to zero gives a step size that takes this coordinate to zero in one step:

$$\alpha = \frac{1}{\lambda_i}.$$

Here that means 0.5 for the first coordinate and 0.05 for the second. No single step size takes both nonzero starting coordinates to zero in one step. A single step size can still make both converge: any positive value below 0.1 does so for this bowl. The steeper coordinate limits the step size. At 0.05 it reaches zero immediately, while the flatter coordinate shrinks gradually. Section 5(a) computes the first step and shows the repeated behavior.

4.2 The best fixed step and the role of the condition number

We now choose a fixed step size for repeated progress. The two fractions left after each step are

$$1 - 2\alpha, \qquad 1 - 20\alpha.$$

We measure progress by the larger magnitude, because that coordinate shrinks more slowly. At 0.05 the magnitudes are 0.9 and zero. Increasing the step size reduces the first fraction, but the second becomes negative and grows in magnitude. The best compromise makes their magnitudes equal. Their signs are opposite there, so

$$\begin{aligned} 1 - 2\alpha &= -(1 - 20\alpha), \\ 1 - 2\alpha &= -1 + 20\alpha, \\ 2 &= 22\alpha, \\ \alpha^* &= \frac{1}{11} \approx 0.0909. \end{aligned}$$

Here α^* names the fixed step size that minimizes the larger remaining fraction. The resulting updates are

$$z_1 \leftarrow \frac{9}{11} z_1, \qquad z_2 \leftarrow -\frac{9}{11} z_2.$$

Each coordinate keeps about 81.8 percent of its magnitude after a step. Because the cost squares both coordinates, it keeps the square of that fraction:

$$f_{\text{next}} = \left(\frac{9}{11}\right)^2 f = \frac{81}{121} f \approx 0.669 f.$$

Starting from cost 49, the first step leaves cost 32.8, and 45 steps take it below 10^{-6}. This criterion concerns the slowest coordinate's repeated progress. It does not minimize the cost after one step from every start: from (3,2), the choice 0.05 gives cost 7.29 after its first step, lower than 32.8.

The Hessian's eigenvalues are the directional curvatures. Their ratio describes the mismatch between the steepest and flattest directions. This is called the **condition number** for a positive definite Hessian:

$$\operatorname{cond}(\mathbf{H}) = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{20}{2} = 10.$$

For a general positive definite quadratic, the same balancing argument uses the smallest and largest curvatures. The following formulas give its best worst-direction factor; intermediate-curvature directions can shrink faster.

The larger contraction factor in (7) is minimized when the two factors are equal in magnitude, $1 - \alpha\lambda_{\min} = -(1 - \alpha\lambda_{\max})$, i.e.

$$\begin{aligned} \alpha^* &= \frac{2}{\lambda_{\min} + \lambda_{\max}}, \\ |\text{factor}| &= \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\operatorname{cond}(\mathbf{H}) - 1}{\operatorname{cond}(\mathbf{H}) + 1}. \end{aligned} \tag{8}$$

The factor bounds how much of the error in each curvature direction remains. For this two-curvature bowl, both directions have the same factor magnitude, so the cost shrinks by its square at every step. For a general positive definite quadratic, that square bounds the remaining cost fraction; the actual cost can fall faster. For $\operatorname{cond}(\mathbf{H}) = 10$ the factor is $9/11 = 0.818$ per step, and 45 steps bring f from 49 below 10^{-6} (89 would be needed if f itself shrank by 0.818). For $\operatorname{cond}(\mathbf{H}) = 100$ the factor is 0.980 and 443 steps are needed; for 10^4, about 44 000. Trajectory optimization problems in week 4 have condition numbers from 10^3 to 10^8, growing like T^4 with the knot count (Deck 05's `komo_two_link.py` prints $\operatorname{cond} \approx 5 \times 10^3$ for the Gauss-Newton matrix at its solution and 2.5×10^5 for its acceleration cost alone at 20 knots); this is why plain gradient descent is not used on them. The larger condition-number counts above describe the two-curvature model or a worst-case bound. α^* minimizes the worst remaining coordinate fraction; the count from a particular start also depends on how much error lies in each direction. A smaller α can have a lower f after a fixed number of steps because it removes z_2 faster (§5 (a): $\alpha = 0.08$ has the lower f at 20 steps).

The arm's bowl. Near a solution of inverse kinematics the residual is small, and $\mathbf{J}^\top\mathbf{J}$ approximates the Hessian of f (§4.5). At the elbow-up solution $(-11.52^\circ, 80.41^\circ)$ its eigenvalues are 0.041 and 0.339, so $\operatorname{cond} = 8.2$, close to this section's bowl; gradient descent at $\alpha^* = 5.26$ needs 56 iterations to a residual of 10^{-6}, and with Armijo 46 (§5 (g)). At the start $(30^\circ, 45^\circ)$, $\mathbf{J}^\top\mathbf{J}$ has eigenvalues 0.015 and 0.495, $\operatorname{cond} = 34 = 5.84^2$, but the true Hessian there is indefinite (eigenvalues -0.017 and 0.339), so the bowl picture holds near a solution and not everywhere.

4.3 Why backtracking terminates, and what Armijo guarantees

For a descent direction, $\nabla f^\top \mathbf{d} < 0$, so the right-hand side of (3) is a line through $f(\mathbf{z})$ with a slightly negative slope $c\nabla f^\top \mathbf{d}$, while the left-hand side is $f(\mathbf{z}) + \alpha\nabla f^\top \mathbf{d} + O(\alpha^2)$, a curve leaving $f(\mathbf{z})$ with the steeper slope $\nabla f^\top \mathbf{d}$. For small enough α the curve is below the line, so halving from any start reaches an accepted α in finitely many trials. Each accepted step then decreases f by at least $c\alpha|\nabla f^\top \mathbf{d}| > 0$, so the method is **monotone**. Backtracking from a large α (from 1, or from the last accepted step grown by 1.2) already avoids needlessly short steps; Wolfe's curvature condition is what quasi-Newton updates need to keep their Hessian estimate positive definite, and it is left to the backup slides. There is no cap on the grown α: α carries units (§3.4) and its natural size differs from problem to problem; a cap at 1, which some implementations use, suits Newton-type directions, whose natural step is 1, and

costs the arm 312 iterations instead of 46 (§5 (g)). With the 1.2-fold growth after each success, the step size finds its own scale: §5 (b) shows it settling between 0.06 and 0.13 for the elongated bowl without anyone tuning it.

What Armijo does not guarantee. Suppose ∇f is L-Lipschitz (on the two-link arm it is, because φ and all its derivatives are bounded). Backtracking by halving from α_0 then accepts some $\alpha \geq \min\{\alpha_0, (1-c)|\nabla f^\top \mathbf{d}|/(L\|\mathbf{d}\|^2)\}$, so each accepted step lowers f by at least

$$c \min\left\{\alpha_0 |\nabla f^\top \mathbf{d}|,\ \frac{(1-c)(\nabla f^\top \mathbf{d})^2}{L\|\mathbf{d}\|^2}\right\}, \qquad \frac{(\nabla f^\top \mathbf{d})^2}{\|\mathbf{d}\|^2} = \cos^2\theta \|\nabla f\|^2. \tag{3'}$$

with θ the angle between $\mathbf{d}$ and $-\nabla f$. The second term does not change when $\mathbf{d}$ is rescaled, so only the angle drives the guarantee to zero; a long $\mathbf{d}$ never lowers the bound, but it makes the α the test needs, about $(1-c)\cos\theta\|\nabla f\|/(L\|\mathbf{d}\|)$, smaller, and in the code that α falls below the fixed floor of 10^{-6}. The code halves from $\alpha = 1$; its last trial is the first α at or below the floor, and when that trial fails the code rejects the direction and stops (`run_gn` reports "stalled"). That is the observed behavior of this implementation and its stopping rule, not a theorem about the direction: at the stall of §5 (k) the trial 2^{-20} fails and the next halving 2^{-21} passes, moving the joints by 8×10^{-4} rad for a cost change of -6×10^{-10}, and floors of 10^{-8} and 10^{-12} reject the direction at Gauss-Newton iterations 10 and 12 with the tip still 9.96 mm from the closest point. The poor progress belongs to the direction, which is within 0.05° of perpendicular to the gradient there; the stopping event belongs to the floor.

4.4 Newton's step from the quadratic model

Minimize the second-order model (1) over $\boldsymbol{\delta}$. Its gradient in $\boldsymbol{\delta}$ is $\nabla f + \mathbf{H}\boldsymbol{\delta}$; setting it to zero,

$$\mathbf{H}\boldsymbol{\delta} = -\nabla f, \boldsymbol{\delta}_{\text{Newton}} = -\mathbf{H}^{-1}\nabla f. \tag{9}$$

For a quadratic $f(\mathbf{z}) = \frac{1}{2}\mathbf{z}^\top \mathbf{A}\mathbf{z} + \mathbf{b}^\top \mathbf{z}$, $\mathbf{H} = \mathbf{A}$ everywhere and $\nabla f(\mathbf{z}^{(i)}) = \mathbf{A}\mathbf{z}^{(i)} + \mathbf{b}$, so $\mathbf{z}^{(i)} + \boldsymbol{\delta} = \mathbf{z}^{(i)} - \mathbf{A}^{-1}(\mathbf{A}\mathbf{z}^{(i)} + \mathbf{b}) = -\mathbf{A}^{-1}\mathbf{b}$ for a symmetric invertible $\mathbf{A}$, the stationary point, **in one step from any start**; it is the minimizer when $\mathbf{A}$ is positive definite (§5 (c)), and a saddle when $\mathbf{A}$ is indefinite (§5 (d)). On a non-quadratic f, Newton is that jump on the bowl fitted at $\mathbf{z}^{(i)}$, repeated; near a minimum with positive definite $\mathbf{H}$ the error after a step is at most a constant times the square of the error before it, quadratic convergence (§5 (f) prints the constants for the arm; "squared" is a rate, not an equality between numbers with units). The direction is invariant to affine rescaling of $\mathbf{z}$, so $\operatorname{cond}(\mathbf{H})$ does not change the number of Newton

steps, while the accuracy of each linear solve in floating point still depends on it. Two problems: $\mathbf{H}$ needs second derivatives of everything, and the step needs a $d \times d$ solve (trivial for 7 joints, and the reason for Deck 05's band structure, where a trajectory of 100 time steps of 7 joints has 700 unknowns); and if $\mathbf{H}$ is not positive definite the step goes to a saddle or a maximum, and may go uphill to get there (§5 (d): $\nabla f^\top \mathbf{d} = +1.5$). The standard fix for the second is to solve $(\mathbf{H} + \rho \mathbf{I})\boldsymbol{\delta} = -\nabla f$ with ρ larger than the magnitude of the most negative eigenvalue of $\mathbf{H}$, so that $\mathbf{H} + \rho\mathbf{I}$ is positive definite; for large ρ the step tends to $-\nabla f / \rho$, a short gradient step. On the arm the exact Hessian is indefinite at the start and Newton walks to a saddle (§5 (i)).

4.5 Gauss-Newton: the least-squares problem of the linearized residual, and its relation to Newton

From the linearized residual. Deck 03's bridge posed $\mathbf{J}\Delta\mathbf{q} \approx \mathbf{e}$ at one guess. In this deck's sign the residual is $\mathbf{r}(\mathbf{q}) = \varphi(\mathbf{q}) - \mathbf{y}^* = -\mathbf{e}$, and at the iterate $\mathbf{q}^{(i)}$ the first-order model of the residual is

$$\mathbf{r}(\mathbf{q}^{(i)} + \boldsymbol{\delta}) \approx \mathbf{r}^{(i)} + \mathbf{J}\boldsymbol{\delta}, \mathbf{r}^{(i)} := \mathbf{r}(\mathbf{q}^{(i)}), \mathbf{J} := \mathbf{J}(\mathbf{q}^{(i)}). \tag{10}$$

The Gauss-Newton step is the minimizer of the least-squares model that (10) defines,

$$m^{(i)}(\boldsymbol{\delta}) = \tfrac{1}{2}\|\mathbf{r}^{(i)} + \mathbf{J}\boldsymbol{\delta}\|^2, \tag{10'}$$

a bowl fitted to the linearized residual at the current iterate. It is not f, and it is rebuilt after every accepted step. Its gradient in $\boldsymbol{\delta}$ is $\mathbf{J}^\top(\mathbf{r}^{(i)} + \mathbf{J}\boldsymbol{\delta})$ (the rule $\partial(\tfrac{1}{2}\mathbf{v}^\top\mathbf{v})/\partial\boldsymbol{\delta} = \mathbf{v}^\top \partial\mathbf{v}/\partial\boldsymbol{\delta}$, transposed), and setting it to zero gives the **normal equations**

$$(\mathbf{J}^\top\mathbf{J})\boldsymbol{\delta} = -\mathbf{J}^\top\mathbf{r}^{(i)}. \tag{12}$$

When $\mathbf{J}$ is square and invertible, as at $\mathbf{q}^{(0)}$ on the two-link arm, (12) gives $\boldsymbol{\delta} = -\mathbf{J}^{-1}\mathbf{r} = \mathbf{J}^{-1}\mathbf{e}$, the step of Deck 03's bridge: the derivation says what that step optimizes. For a general residual $\mathbf{r}(\mathbf{z})$ with Jacobian $\mathbf{J}_r$ the same three lines hold with $\mathbf{J}_r$ in place of $\mathbf{J}$.

Why it is Newton without second derivatives. With $f = \frac{1}{2}\sum_i r_i^2$, differentiate term by term: $\nabla f = \sum_i r_i \nabla r_i$, and the rows of $\mathbf{J}_r$ are the $\nabla r_i^\top$, so

$$\nabla f = \mathbf{J}_r^\top \mathbf{r}. \tag{11a}$$

Differentiate once more with the product rule, $\nabla(r_i \nabla r_i) = \nabla r_i \nabla r_i^\top + r_i \nabla^2 r_i$:

$$\mathbf{H} = \sum_i \nabla r_i \nabla r_i^\top + \sum_i r_i \nabla^2 r_i = \mathbf{J}_r^\top \mathbf{J}_r + \sum_{i=1}^{m} r_i \nabla^2 r_i. \tag{11}$$

Newton solves $\mathbf{H}\boldsymbol{\delta} = -\nabla f$; (12) solves the same system with the residual-curvature sum of (11) dropped. So "drop the sum" is a consequence of linearizing the residual, not a recipe. The dropped sum is small when the residuals are small at the solution (every $r_i \approx 0$) or when $\mathbf{r}$ is nearly linear (every $\nabla^2 r_i \approx 0$). The small-residual condition holds near a reachable regular solution ($\mathbf{J}(\mathbf{q}^*)$ of full rank); the residual's second derivatives need not vanish. Under the regularity assumptions of §4.10, Gauss–Newton retains the quadratic rate there. On the reach boundary the residual still vanishes but $\mathbf{J}(\mathbf{q}^*)$ is singular, and the rate is only linear (§5 (j)). In the unreachable two-link example, the residual stays nonzero and its curvature term remains relevant. The linear model is also trusted near a singular configuration (§5(k)).

For inverse kinematics $\mathbf{r}(\mathbf{q}) = \varphi(\mathbf{q}) - \mathbf{y}^*$ and $\mathbf{J}_r = \mathbf{J}$, so $\nabla f = \mathbf{J}^\top(\varphi(\mathbf{q}) - \mathbf{y}^*)$: this is the transpose rule of essentials §6 applied to $L = \frac{1}{2}\|\mathbf{x} - \mathbf{y}^*\|^2$ with $\mathbf{x} = \varphi(\mathbf{q})$, and §5 (e) checks it against central differences. The essentials' residual is $\mathbf{e} = \mathbf{y}^* - \varphi = -\mathbf{r}$, so their step $\mathbf{J}^+\mathbf{e}$ and this deck's $-(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top\mathbf{r}$ are the same step written with opposite signs.

What positive semidefinite buys, and what it does not. $\mathbf{J}_r^\top\mathbf{J}_r$ is positive semidefinite ($\boldsymbol{\delta}^\top\mathbf{J}_r^\top\mathbf{J}_r\boldsymbol{\delta} = \|\mathbf{J}_r\boldsymbol{\delta}\|^2 \geq 0$), so the model bowl never curves downward, and the step never climbs because of negative model curvature the way Newton's does in §5 (d) and (i); at $\mathbf{q}^{(0)}$ the exact Hessian has eigenvalues $-0.017, 0.339$ while $\mathbf{J}^\top\mathbf{J}$ has $0.015, 0.495$. That reduces Newton's saddle problem; it does not remove it, because the method is still local: at a stationary point of f the right-hand side $-\mathbf{J}_r^\top\mathbf{r}$ is zero. The damped and minimum-norm Gauss–Newton choices are then zero, whether or not the point is a minimum (§5 (p): a reachable target at which the stretched arm is a saddle of f). Any solution of the normal equations satisfies

$$\nabla f^\top\boldsymbol{\delta} = -\boldsymbol{\delta}^\top\mathbf{J}_r^\top\mathbf{J}_r\boldsymbol{\delta} = -\|\mathbf{J}_r\boldsymbol{\delta}\|^2 \leq 0.$$

At a nonstationary point, this quantity is strictly negative, so the proposed direction allows a line search. At the square, invertible starting Jacobian of §5(f), it is $-\|\mathbf{r}\|^2 = -0.0897$. The familiar inverse formula requires full column rank:

$$\nabla f^\top\boldsymbol{\delta} = -\mathbf{r}^\top\mathbf{J}_r(\mathbf{J}_r^\top\mathbf{J}_r)^{-1}\mathbf{J}_r^\top\mathbf{r}.$$

What remains is rank: $\mathbf{J}_r^\top\mathbf{J}_r$ is singular exactly when $\mathbf{J}_r$ lacks full column rank, at $\det\mathbf{J} = 0$ on the two-link arm and at every $\mathbf{q}$ on a redundant arm with $n > m$. Then the inverse formula $(\mathbf{J}_r^\top\mathbf{J}_r)^{-1}\mathbf{J}_r^\top$ fails, but (12) is still consistent ($\mathbf{J}_r^\top\mathbf{r}$ lies in the range of $\mathbf{J}_r^\top\mathbf{J}_r$) and the linear least-squares problem still has minimizers: the pseudo-inverse picks the one of minimum norm, and damping (§4.6) makes the step unique.

4.6 Levenberg–Marquardt: damping as regularization, and the two limits

Some joint-step directions change the linearized residual more than others. Their gains per unit joint-step norm are called **singular values**. For the selected units and scaling, a singular value σ_i measures the linearized residual change produced by a unit joint step along its corresponding direction. With position residuals in meters and joint steps in radians, these gains have units of meters per radian. The nonzero gains describe directions orthogonal to any null space. A small gain identifies a direction in which the inverse step can become large. The gains change if we change the units or scaling. We use them to explain damping below; their matrix factorization is optional in Deck 03.

$$(\mathbf{J}_r^\top \mathbf{J}_r + \rho \mathbf{I})\boldsymbol{\delta} = -\mathbf{J}_r^\top \mathbf{r}, \rho > 0. \tag{13}$$

Two readings of (13). It is the Newton step for the modified cost $\frac{1}{2}\|\mathbf{r}(\mathbf{z}) + \mathbf{J}_r\boldsymbol{\delta}\|^2 + \frac{1}{2}\rho\|\boldsymbol{\delta}\|^2$, i.e. Gauss-Newton with a penalty on the step length: the same algebra as the damped least squares of essentials §7, with the residual in place of the velocity. And it interpolates: as $\rho \to 0$ it is Gauss-Newton; as $\rho \to \infty$, $\boldsymbol{\delta} \approx -\mathbf{J}_r^\top \mathbf{r}/\rho = -\nabla f/\rho$, a gradient step with step size $1/\rho$. Damping makes the linear system nonsingular for every $\mathbf{J}_r$ (each eigenvalue of $\mathbf{J}_r^\top \mathbf{J}_r + \rho\mathbf{I}$ is at least ρ), so the step exists, is unique and stays bounded at rank-deficient $\mathbf{J}_r$ (its length is at most $\|\mathbf{r}\|/(2\sqrt{\rho})$, because $\sigma/(\sigma^2+\rho) \le 1/(2\sqrt{\rho})$ for every singular value), and the cost is slower convergence: on a linear residual the error after a step is $\rho\big(\mathbf{J}_r\mathbf{J}_r^\top + \rho\mathbf{I}\big)^{-1}$ times the error before it, so along a singular direction of $\mathbf{J}_r$ with singular value σ_i the error shrinks by the factor $\rho/(\sigma_i^2+\rho)$ per step, linear convergence with a rate set by the damping and the weakest direction. That factor is exact for a fixed linear residual map and a good approximation near a regular zero-residual solution; it is not a per-step rate for the nonlinear arm far from a solution, and at a nonzero-residual optimum the dropped term of (11) changes the picture. Where it applies it predicts the counts: the two-link arm's 21 and 165 iterations at $\rho = 0.05$ and 0.5 are this factor with $\sigma_{\min} = 0.20$ (§5 (g)), and the Panda's 18 steps are this factor with $\sigma_{\min} = 0.026$ and $\rho = 10^{-3}$ (§5 (o)).

Fixed and adaptive damping. Holding ρ fixed gives damped Gauss-Newton; Levenberg–Marquardt methods adapt ρ, and the course's simple rule is one of them: after a step that lowers f, divide ρ by 10; otherwise multiply it by 10 and try the step again (production codes use trust-region rules, and the name covers all of them). On the arm the adaptive rule from $\rho = 0.05$ reaches $(0.5, 0.2)$ in four steps with the residual falling at every step $(0.12, 9.4\times10^{-3}, 8.7\times10^{-5}, 1.2\times10^{-7})$; from $\rho = 10^{-3}$ it also settles the unreachable target of §5 (k) (step below 10^{-6} from step 20, stationary from step 24, printed) and the Panda's unreachable target of §5 (o), where one fixed ρ does not serve both. The fixed-damping and adaptive-damping runs below distinguish these two methods.

4.7 The damped inverse-kinematics step, line by line

Start from (5′) with a general $\mathbf{K}$ and $\mathbf{D}$, at the iterate $\mathbf{q}^{(i)}$, and linearize the forward kinematics there:

$$\varphi(\mathbf{q}) \approx \mathbf{y}^{(i)} + \mathbf{J}\boldsymbol{\delta}, \mathbf{y}^{(i)} := \varphi(\mathbf{q}^{(i)}), \mathbf{J} := \mathbf{J}(\mathbf{q}^{(i)}), \boldsymbol{\delta} := \mathbf{q} - \mathbf{q}^{(i)}. \quad (14)$$

Substitute (14) into (5′), and call the result $m^{(i)}(\boldsymbol{\delta})$, the local model of the step. It is not f, and it is rebuilt after every accepted step; it is (10′) with the weights $\mathbf{K}$ and $\mathbf{D}$ carried along:

$$m^{(i)}(\boldsymbol{\delta}) = \tfrac{1}{2}\|\mathbf{y}^{(i)} - \mathbf{y}^* + \mathbf{J}\boldsymbol{\delta}\|^2_{\mathbf{K}} + \tfrac{1}{2}\|\boldsymbol{\delta}\|^2_{\mathbf{D}}. \quad (15)$$

Differentiate $m^{(i)}$ with respect to $\boldsymbol{\delta}$ (the rule $\partial(\frac{1}{2}\mathbf{v}^\top\mathbf{M}\mathbf{v})/\partial\boldsymbol{\delta} = \mathbf{v}^\top\mathbf{M}\partial\mathbf{v}/\partial\boldsymbol{\delta}$ for symmetric $\mathbf{M}$):

$$\partial m^{(i)}/\partial\boldsymbol{\delta} = (\mathbf{y}^{(i)} - \mathbf{y}^* + \mathbf{J}\boldsymbol{\delta})^\top\mathbf{K}\mathbf{J} + \boldsymbol{\delta}^\top\mathbf{D}. \quad (16)$$

Set it to zero, transpose, and collect the terms containing $\boldsymbol{\delta}$ on one side:

$$\begin{aligned} &\mathbf{J}^\top\mathbf{K}(\mathbf{y}^{(i)} - \mathbf{y}^*) + \mathbf{J}^\top\mathbf{K}\mathbf{J}\boldsymbol{\delta} + \mathbf{D}\boldsymbol{\delta} = 0 \\ \Longrightarrow\ &(\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})\boldsymbol{\delta} = \mathbf{J}^\top\mathbf{K}(\mathbf{y}^* - \mathbf{y}^{(i)}). \end{aligned} \quad (17)$$

$\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D}$ is positive definite whenever $\mathbf{D}$ is (or whenever $\mathbf{D} = 0$ and $\mathbf{J}$ has full column rank), so it can be inverted:

$$\mathbf{q}^{(i+1)} = \mathbf{q}^{(i)} + \mathbf{J}^\sharp(\mathbf{y}^* - \mathbf{y}^{(i)}), \mathbf{J}^\sharp := (\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})^{-1}\mathbf{J}^\top\mathbf{K}. \quad (18)$$

The **Woodbury identity** gives an equivalent form when $\mathbf{D}$ and $\mathbf{K}$ are invertible:

$$\mathbf{J}^\sharp = \mathbf{D}^{-1}\mathbf{J}^\top(\mathbf{J}\mathbf{D}^{-1}\mathbf{J}^\top + \mathbf{K}^{-1})^{-1}. \quad (19)$$

This form inverts an $m \times m$ matrix instead of an $n \times n$ one and is the cheaper form for a redundant arm ($n = 7 > m = 3$). The code checks (19) against (18) for the two-link arm: they agree to 1.3×10^{-15} (§5 (n)). Because (14) is only a linearization, (18) is applied **iteratively**, re-linearizing at each new iterate:

$$\mathbf{q}^{(i+1)} = \mathbf{q}^{(i)} + \mathbf{J}^\sharp(\mathbf{q}^{(i)})(\mathbf{y}^* - \varphi(\mathbf{q}^{(i)})). \quad (20)$$

When to stop. A run reports both its task error and its gradient norm:

$$\|\varphi(\mathbf{q}) - \mathbf{y}^*\|, \qquad \|\nabla f\| = \|\mathbf{J}^\top\mathbf{K}(\varphi(\mathbf{q}) - \mathbf{y}^*)\|.$$

The supplied solver distinguishes four outcomes:

- A run is **solved** when its residual is below the residual tolerance.

- A run is **stationary** when its residual is larger, but it passes the relative gradient test below. It may have reached a least-squares minimum, as in §5(k), or a saddle, as in §5(p).
- A run is **stalled** when Armijo rejects the trial at the step-size floor. This does not establish stationarity (§4.3 and §5(k)).
- A run reaches its **iteration limit** when it uses the permitted number of iterations without another outcome. The code returns the literal string `limit`.

For an unweighted cost, the course's relative gradient test is

$$\|\mathbf{J}^\top \mathbf{r}\| \le \varepsilon_g \|\mathbf{J}\|_F \|\mathbf{r}\|, \qquad \varepsilon_g = 10^{-8}, \qquad \mathbf{K} = \mathbf{I}.$$

The Frobenius norm combines the squared entries of the Jacobian. For a weighted cost, apply the same test to the residual and Jacobian transformed by $\mathbf{K}^{1/2}$. Multiplying the whole cost by a positive constant then changes neither side of the ratio. An absolute gradient threshold would stop the boundary run of §5(j) early, at a residual of $5.5 \times 10^{-6}\ \mathrm{m}$, because the gradient shrinks faster than the residual at that singular solution. The relative test lets that run finish. At the stretched arm of §5(p), the gradient is exactly zero even though the target is reachable and the error remains.

A small-step stop is optional. It compares the accepted step norm to a tolerance and appears in the slides' gradient-descent algorithm. It is off in every run of §5. Reported steps below $10^{-6}\ \mathrm{rad}$ are therefore observations rather than termination events. The code also returns `diverged` if a step is nonfinite or exceeds $10^{3}\ \mathrm{rad}$. The unprotected run of §5(k) triggers this check at iteration 364.

No outcome proves that a target is unreachable. For the two-link arm, geometry establishes that result when the target radius lies outside its reachable annulus:

$$\|\mathbf{y}^*\| \notin [|l_1 - l_2|, l_1 + l_2].$$

Report the outcome and both norms to state what the numerical run established.

Optional reference on the gradient test. This test is not an angle to the range of $\mathbf{J}$. It also differs from MINPACK's largest columnwise cosine criterion,

$$\max_j \frac{|\mathbf{J}_j^\top \mathbf{r}|}{\|\mathbf{J}_j\| \|\mathbf{r}\|}.$$

Here $\mathbf{J}_j$ is column j of the Jacobian. That criterion is stricter for a weak column. Our test can pass with error left along such a direction. For example,

$$\mathbf{J} = \mathrm{diag}(1, 10^{-9}), \qquad \mathbf{r} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The course's ratio is approximately 10^{-9}, so the test passes, although the residual is parallel to the second column. This is a limitation of the stopping rule, and it is another reason to report task error as well as the gradient.

4.8 Four special cases of (18), and what they connect to

- **Square, invertible $\mathbf{J}$, $\mathbf{D} = 0$:** $\mathbf{J}^\sharp = (\mathbf{J}^\top\mathbf{K}\mathbf{J})^{-1}\mathbf{J}^\top\mathbf{K} = \mathbf{J}^{-1}\mathbf{K}^{-1}\mathbf{J}^{-\top}\mathbf{J}^\top\mathbf{K} = \mathbf{J}^{-1}$. One step of (20) is Deck 03's differential inverse applied to the remaining displacement instead of a velocity: Deck 03 multiplied a requested velocity in m/s and integrated the rates over Δt (its slide 44); here there is no time step, and the step is repeated with a fresh $\mathbf{J}$. The second column of $\mathbf{J}^{-1}$ at $(30^\circ, 45^\circ)$ times 0.1 m/s is Deck 03's $(0.3415, -0.5772)$ rad/s. This is (12) applied to the weighted residual $\mathbf{r} = \mathbf{K}^{1/2}(\varphi - \mathbf{y}^*)$ and its Jacobian $\mathbf{J}_r = \mathbf{K}^{1/2}\mathbf{J}$.
- **$\mathbf{K} = w\mathbf{I}$, $w \to \infty$ (a hard, isotropically weighted task), $\mathbf{D} = \mathbf{I}$:** by (19), $\mathbf{J}^\sharp \to \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$, the closed form of the **pseudo-inverse $\mathbf{J}^+$** at full row rank: the minimum-norm joint change that achieves the task error exactly. The limit itself is $\mathbf{J}^+$ at every rank (when the rank is deficient it drops the unreachable component, essentials §7, with the numbers in §8); only the closed form $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$ needs full row rank. At full row rank the shape of the weight does not matter: by (19) with $\mathbf{D} = \mathbf{I}$, $\mathbf{J}^\sharp = \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top + \mathbf{K}^{-1})^{-1}$, and $\mathbf{K}^{-1} \to 0$ for any $\mathbf{K} = w \cdot \mathbf{K}_0$ with $\mathbf{K}_0 \succ 0$. When $\mathbf{J}$ lacks full row rank the weight must be isotropic: with $\mathbf{J} = (1, 1)^\top$ (rank 1, two task rows) and $\mathbf{K} = t \cdot \mathrm{diag}(1, 4)$ the limit is the row $(0.2, 0.8)$, not the pseudo-inverse's $(0.5, 0.5)$, because an anisotropic $\mathbf{K}$ changes how the unreachable residual is projected. It is the redundant-arm default when far from singularities.
- **$\mathbf{K} = \mathbf{I}$, $\mathbf{D} = \rho\mathbf{I}$:** $\mathbf{J}^\sharp = (\mathbf{J}^\top\mathbf{J} + \rho\mathbf{I})^{-1}\mathbf{J}^\top$, exactly the damped step (13) with $\mathbf{r} = \varphi - \mathbf{y}^*$ and $\mathbf{J}_r = \mathbf{J}$, since $-\mathbf{J}_r^\top\mathbf{r} = \mathbf{J}^\top(\mathbf{y}^* - \varphi)$. This is the form the code and the module use.
- **$\mathbf{D}$ as a preference:** any positive definite $\mathbf{D}$ says "changes in these joints are cheaper than in those"; KOMO weights its features in the same way, through `scale`, and not joints (4.2).

4.9 Damping preserves the set of fixed points

A fixed point of (20) has a zero proposed step. Substituting the definition of $\mathbf{J}^\sharp$ from (18) gives

$$\begin{aligned} \mathbf{J}^\sharp(\mathbf{y}^* - \varphi(\mathbf{q})) &= 0, \\ (\mathbf{J}^\top\mathbf{K}\mathbf{J} + \mathbf{D})^{-1}\mathbf{J}^\top\mathbf{K}(\mathbf{y}^* - \varphi(\mathbf{q})) &= 0, \\ \mathbf{J}^\top\mathbf{K}(\mathbf{y}^* - \varphi(\mathbf{q})) &= 0. \end{aligned}$$

The equivalence requires $\mathbf{J}^\top\mathbf{K}\mathbf{J}+\mathbf{D}$ to be invertible. This holds when $\mathbf{D}$ is positive definite, or when $\mathbf{D}=0$ and $\mathbf{J}$ has full column rank. The right-hand condition is the stationarity condition of (5). It contains no step weight $\mathbf{D}$, so the **set** of fixed points is the same for every ρ; which one the iteration reaches depends on the start and on ρ. From $(30^\circ, 45^\circ)$, $\rho=0$ and $\rho=0.5$ agree to 4×10^{-6} rad, but on a 35×35 grid of starts they reach different solutions from 384 of the 1225 starts (§5 (l)); on a redundant arm, different ρ can end at different $\mathbf{q}$.

The essentials' sentence "increasing ρ accepts more tracking error" (Appendix C there) and this section agree, because the accuracy it gives up applies to one step only. Later steps can reduce the leftover error. With $\rho=0.05$ the first step's linear model leaves a predicted residual of $0.147\ \mathrm{m}$ (against 0 undamped; $0.217\ \mathrm{m}$ at $\rho=0.5$), the true residual after the step is $0.115\ \mathrm{m}$ (against $0.174\ \mathrm{m}$ undamped), and the next step asks for the leftover. When a run converges to a zero-residual solution, that solution is exact whatever ρ is. From $\mathbf{q}^{(0)}=(30^\circ,45^\circ)$, damping adds iterations (21 instead of 5) and ends at the same solution (§5 (g)); from other starts it can end at the other solution, and it can also succeed where the undamped step fails: on the grid of §5 (l), $\rho=0$ fails from 27 starts and $\rho=0.5$ from none. Reachability alone does not guarantee a zero-residual outcome, because a reachable target can hold a stationary point at which every damped step is zero (§5 (p)). A fixed posture $\mathbf{q}_0$ in place of the moving iterate is a different problem and does change the answer (§3.4).

4.10 Quadratic convergence at a regular solution and linear convergence at the stretched arm

For the two-link position task, near a solution $\mathbf{q}^*$ with $\varphi(\mathbf{q}^*)=\mathbf{y}^*$ and $\mathbf{J}(\mathbf{q}^*)$ of full rank, the neglected term in (11) is $\sum r_i\nabla^2 r_i\to0$ because $\mathbf{r}\to0$; Gauss-Newton therefore behaves like Newton with the exact Hessian, and the error after a step is a constant times the square of the error before it (§5 (f): residual exponents $-1,-1,-2,-3,-6,-11$ for $i=0\ldots5$, quadratic from step 2 on, with the constants $1.19, 1.18, 1.72, 1.30$). When the solution is singular ($\det\mathbf{J}(\mathbf{q}^*)=0$, the stretched arm at the reach boundary), the linear model (14) is first-order accurate along the arm's circumferential direction but has no radial component, while the true φ falls short of the stretched arm's reach at second order. The shortfall is $l_1+l_2-\|\varphi(\mathbf{q})\|\approx l_1l_2q_2^2/(2(l_1+l_2))$ for small q_2 (exactly $2l_1l_2(1-\cos q_2)/(l_1+l_2+\|\varphi(\mathbf{q})\|)$), which is $3.54\times10^{-4}m$ at $q_2=3.68^\circ$. Why q_2 halves: along the null direction of $\mathbf{J}$ at the solution the residual is quadratic in the offset, and a Gauss-Newton step on a residual x^2 with Jacobian $2x$ is $x-x^2/(2x)=x/2$, Newton's behavior on a double root. The iterates approach the solution along the direction $(l_2,-(l_1+l_2))$, so $q_1/q_2\to-0.43$, as the table of §5 (j) shows. Since this radial error is quadratic in q_2, halving q_2 divides it by four: one step removes three quarters of it, which is the factor of four in the residuals of §5 (j). When the target is unreachable, the residual at the least-squares solution is not zero, the dropped term is

not small, and at that solution, the stretched arm, $\mathbf{J}^\top\mathbf{J}$ is singular: the inverse formula fails while the least-squares subproblem keeps its minimizers, and the undamped iteration cannot settle, which is §5 (k).

5. Worked examples

All numbers are printed by `code/descent.py` ((a)–(d)), `code/gauss_newton_ik.py` ((e)–(n) and (p), the letters of its printout) and `drake/02-gauss-newton-ik/gn_ik_panda.py` ((o)), and are rounded from the printed digits. Conventions: the cost carries the $\frac{1}{2}$; Armijo uses $c = 0.01$, halves from $\alpha = 1$, and rejects a direction when its first trial at or below the floor 10^{-6} fails (the run then reports "stalled"; a smaller trial may still pass, (k)); on the Gauss-Newton direction the search restarts from $\alpha = 1$ at every iteration, because the natural length of a Newton-type step is 1, while gradient descent grows α by 1.2 after each accepted step with no cap; every run ends with one of the outcomes of §4.7, and "solved" means $\|\varphi - \mathbf{y}^*\| < 10^{-6}$.

(a) Gradient descent on $f = z_1^2 + 10z_2^2$ from $\mathbf{z}^{(0)} = (3, 2)$, $f(\mathbf{z}^{(0)}) = 49$, $\nabla f(\mathbf{z}^{(0)}) = (6, 40)$, $\mathbf{H} = \mathrm{diag}(2, 20)$, $\mathrm{cond}(\mathbf{H}) = 10$. First substitute two step sizes into (7):

α	factor on z_1	factor on z_2	first point
0.05	$1 - 2(0.05) = 0.9$	$1 - 20(0.05) = 0$	$(0.9 \cdot 3, 0 \cdot 2) = (2.7, 0)$
0.5	$1 - 2(0.5) = 0$	$1 - 20(0.5) = -9$	$(0 \cdot 3, -9 \cdot 2) = (0, -18)$

The first choice takes z_2 to zero and moves z_1 closer to zero. The second takes z_1 to zero but sends z_2 past zero and farther away. With a fixed step size, we repeat the same coordinate multipliers at every iteration. The following table compares the result after twenty steps:

α	factors $(1 - 2\alpha, 1 - 20\alpha)$	$f(\mathbf{z}^{(20)})$	behavior
0.02	$(0.96, 0.60)$	1.758	converges, slowly in z_1
0.05	$(0.90, 0.00)$	0.133	z_2 solved in one step; z_1 crawls
0.08	$(0.84, -0.60)$	0.0084	z_2 zigzags with decaying amplitude

α	factors $(1-2\alpha, 1-20\alpha)$	$f(\mathbf{z}^{(20)})$	behavior
0.0909	$(0.818, -0.818)$	0.0159	the best fixed step, (8)
0.10	$(0.80, -1.00)$	40.0	z_2 oscillates between ± 2 forever
0.11	$(0.78, -1.20)$	58791	diverges

At $\alpha^* = 0.0909$, $f < 10^{-6}$ takes 45 steps. Figure `fig/gd_steps.png`.

(b) Armijo backtracking at $\mathbf{z}^{(0)}$ with $\mathbf{d} = -\nabla f = (-6, -40)$, $\nabla f^\top \mathbf{d} = -1636$, $c = 0.01$, so the acceptance threshold is $49 - 16.36\alpha$:

trial α	$f(\mathbf{z}^{(0)} + \alpha \mathbf{d})$	threshold	
1	14449	32.64	reject
0.5	3240	40.82	reject
0.25	642.25	44.91	reject
0.125	95.06	46.96	reject
0.0625	9.39	47.98	**accept**

$\mathbf{z}^{(1)} = (2.625, -0.5)$, $f = 9.39$. Continuing with the grow-by-1.2 rule the accepted steps are $0.0625, 0.075, 0.09, 0.108, 0.1296, 0.0778, \ldots$: the search hovers around the best fixed step without knowing it, and $f(\mathbf{z}^{(12)}) = 0.098$. Figure `fig/armijo.png`.

(c) Newton on the same bowl: $\mathbf{z}^{(0)} - \mathbf{H}^{-1}\nabla f(\mathbf{z}^{(0)}) = (3, 2) - (6/2, 40/20) = (0, 0)$. One step.

(d) Newton on a saddle. $f = z_1^2 - z_2^2$, $\mathbf{z} = (0.5, 1)$, $f = -0.75$, $\nabla f = (1, -2)$, $\mathbf{H} = \operatorname{diag}(2, -2)$: the Newton step $\mathbf{d} = (-0.5, -1)$ lands on $(0, 0)$, the saddle, and it climbs to get there, $\nabla f^\top \mathbf{d} = +1.5 > 0$ and f rises from -0.75 to 0. With $(\mathbf{H} + \rho \mathbf{I})$, $\rho = 1$ is not enough ($\mathbf{H} + \mathbf{I}$ still has a negative eigenvalue and the direction climbs, $\nabla f^\top \mathbf{d} = +3.7$); with $\rho = 3 > 2$ the direction is $(-0.2, 2)$ and $\nabla f^\top \mathbf{d} = -4.2 < 0$, a descent direction.

(e) One gradient step on the arm (session 3.1). $\mathbf{q}^{(0)} = (30^\circ, 45^\circ)$, $\varphi(\mathbf{q}^{(0)}) = (0.4241, 0.4898)$, $\mathbf{y}^* = (0.5, 0.2)$ ($\|\mathbf{y}^*\| = 0.5385$, inside the annulus), $f(\mathbf{q}^{(0)}) = 0.0449\ \mathrm{m}^2$. The gradient by (11a) is $\nabla f(\mathbf{q}^{(0)}) = \mathbf{J}^\top(\varphi - \mathbf{y}^*) = (0.1601, 0.0445)\ \mathrm{m}^2/\mathrm{rad}$; central differences with $h = 10^{-6}$ give the same to 9×10^{-12}. One gradient step with Armijo from $\alpha = 1$ is accepted at once: the joints

move by $(-9.17^\circ, -2.55^\circ)$ to $(20.83^\circ, 42.45^\circ)$, the tip from $(0.4241, 0.4898)$ to $(0.5088, 0.4102)$, and the residual from $0.2996\ \mathrm{m}$ to $0.2104\ \mathrm{m}$. This is Jacobian-transpose inverse kinematics. The Gauss-Newton step of (f) from the same point is $(-52.7^\circ, 74.1^\circ)$, 70° away from $-\nabla f$: the gradient points mostly across the valley of `fig/ik_iterates.png`, and the direction, not the step size, is what the second session fixes.

(f) The Gauss-Newton step by hand, then the iteration. First step of (20) with $\mathbf{K} = \mathbf{I}$, $\mathbf{D} = 0$:

$$\mathbf{J}(\mathbf{q}^{(0)}) \approx \begin{bmatrix} -.4898 & -.2898 \\ .4241 & .0776 \end{bmatrix}, \qquad \mathbf{y}^* - \varphi(\mathbf{q}^{(0)}) \approx \begin{bmatrix} .0759 \\ -.2898 \end{bmatrix}.$$

$$\mathbf{J}^\sharp = \mathbf{J}^{-1} \approx \begin{bmatrix} .9151 & 3.4151 \\ -4.9975 & -5.7721 \end{bmatrix},$$

$$\boldsymbol{\delta} = \mathbf{J}^\sharp(\mathbf{y}^* - \varphi(\mathbf{q}^{(0)})) \approx \begin{bmatrix} -.9201 \\ 1.2931 \end{bmatrix}\ \mathrm{rad} = \begin{bmatrix} -52.7^\circ \\ 74.1^\circ \end{bmatrix}.$$

Armijo accepts this step at $\alpha = 1$: f falls from 0.0449 to 0.0151, the threshold is 0.0440 (one percent of the predicted drop 0.0897), and the test puts no limit on the step's length. The linear model promised a residual of 0 and delivered $0.174\ \mathrm{m}$.

i	$\mathbf{q}^{(i)}$ (deg)	$\varphi(\mathbf{q}^{(i)})$	$\lVert\varphi - \mathbf{y}^*\rVert$	$\lVert\mathbf{s}\rVert$ (rad)
0	$(30.00, 45.00)$	$(0.4241, 0.4898)$	3.0×10^{-1}	
1	$(-22.72, 119.09)$	$(0.3357, 0.1437)$	1.7×10^{-1}	1.587
2	$(-16.53, 84.53)$	$(0.4959, 0.1643)$	3.6×10^{-2}	0.613
3	$(-11.55, 80.73)$	$(0.4985, 0.2003)$	1.5×10^{-3}	0.109
4	$(-11.52, 80.41)$	$(0.5000, 0.2000)$	4.0×10^{-6}	0.006
5	$(-11.52, 80.41)$	$(0.5000, 0.2000)$	2.0×10^{-11}	0.000

Residual exponents $-1, -1, -2, -3, -6, -11$: quadratic convergence from step 2 on; the ratios $\mathrm{res}_{i+1}/\mathrm{res}_i^2$ are $1.19, 1.18, 1.72, 1.30$ for $i = 1 \ldots 4$, a constant times the square and not the square itself. The first step overshoots (the elbow swings 74.1° and the shoulder $-52.7°$, a step of 1.59 rad, for a 30 cm error) because the linearization is trusted far from $\mathbf{q}^{(0)}$; damping, not the line search, is what shortens it (g).

(g) Damping, gradient descent, and the arm's bowl. With $\rho = 0.05$ the first step is $(-23.8^\circ, 11.5^\circ)$ and leaves $0.115\ \mathrm{m}$ (the linear model predicted $0.147\ \mathrm{m}$ left), with the elbow at 56.5° instead of 119°; with $\rho = 0.5$ it is $(-9.7^\circ, -1.4^\circ)$, close to the gradient step of (e), and leaves $0.208\ \mathrm{m}$. Both reach the same fixed point, $\rho =$

0.05 in 21 iterations and $\rho = 0.5$ in 165: the rate $\rho/(\sigma_{\min}^2 + \rho)$ of §4.6 with $\sigma_{\min} = 0.20$. Gradient descent with Armijo on the same cost needs 46 iterations (with the former cap $\alpha \le 1$ it needed 312, because α wants to be about 5 here, §3.4). Near the solution the cost is a bowl with Hessian $\mathbf{J}^\top\mathbf{J}$, eigenvalues 0.041 and 0.339, $\mathrm{cond} = 8.2$; the best fixed step $\alpha^* = 5.26$ needs 56 iterations. Figure `fig/ik_iterates.png`: left, the cost over joint space with the two valleys, the saddle $(21.8^\circ, 0^\circ)$ between them, the five Gauss-Newton iterates and the 46 gradient-descent iterates; right, the residual per iteration for the four runs.

(h) The other solution. From the elbow-down start $\mathbf{q}^{(0)} = (60^\circ, -45^\circ)$ Gauss-Newton converges in 4 iterations to $(55.12^\circ, -80.41^\circ)$. Gauss-Newton is a local method, so the start decides which solution it reaches.

(i) Exact Newton on the arm. With the second derivatives of φ included (the exact $\mathbf{H}$ of (11)), $\mathbf{H}(\mathbf{q}^{(0)})$ is indefinite (eigenvalues -0.017, 0.339), and Newton from $\mathbf{q}^{(0)}$ walks in five steps to $(21.80^\circ, 0^\circ)$: the arm straight, pointing at the target and past it (tip at $(0.650, 0.260)$, $0.700\ \mathrm{m}$ from the base, which is $l_1 + l_2$, target at $0.539\ \mathrm{m}$, so the residual $0.1615\ \mathrm{m}$ is $0.700 - 0.539$), $\nabla f = 0$, residual $0.1615\ \mathrm{m}$, Hessian eigenvalues -0.024 and 0.442. It is the saddle between the two valleys of the figure. Gauss-Newton from the same start reaches the target in five steps. Its matrix $\mathbf{J}^\top\mathbf{J}$ is never indefinite, so from this start it never climbs toward the saddle the way Newton does. The saddle is still a stationary point of f, where $\mathbf{J}^\top\mathbf{r} = 0$ and the Gauss-Newton step is zero, so the sum-of-squares structure reduces the saddle problem and does not remove it (§4.5, §5 (p)). The comparison depends on the start: toward the unreachable $(0.8, 0.1)$ exact Newton from $\mathbf{q}^{(0)}$ ends at the folded-arm stationary point $(7.1^\circ, -180^\circ)$ with residual $0.706\ \mathrm{m}$, while from Gauss-Newton's second iterate $(7.6^\circ, 0.8^\circ)$ it reaches the closest reachable point in two steps (printed by (i)). Adaptive damping from $\rho = 0.05$ (§4.6) reaches the target in four steps with the residual falling at every step ($0.12, 9.4 \times 10^{-3}, 8.7 \times 10^{-5}, 1.2 \times 10^{-7}$, printed by (g)).

(j) A target on the reach boundary, $\mathbf{y}^* = (0.7, 0)$. Gauss-Newton converges, but only linearly: q_2 halves per step and the residual, quadratic in q_2 near the stretched arm, is divided by four per step: $3.8 \times 10^{-1}, 1.3 \times 10^{-1}, 2.5 \times 10^{-2}, 5.7 \times 10^{-3}, 1.4 \times 10^{-3}, 3.5 \times 10^{-4}, \ldots$, "solved" after eleven iterations at 3.5×10^{-7}, with q_2 halving toward 0 each time ($14.8^\circ, 7.4^\circ, 3.7^\circ, 1.8^\circ, \ldots$). The solution is the stretched arm, $\det \mathbf{J} = l_1 l_2 \sin q_2 = 0$, and §4.10 explains the factor 0.5. An absolute gradient test would have stopped this run early: at iteration 9 the gradient is 8.6×10^{-9} while the residual is still 5.5×10^{-6}, because $\mathbf{J}^\top\mathbf{r}$ shrinks faster than $\mathbf{r}$ at a singular solution; the relative gradient test of §4.7 lets it finish (printed by (j)). Armijo on the same direction accepts every full step, so it prints the same table and

the same word. Damping is slow here: $\rho = 0.01$ with full steps has residual 6×10^{-4} after 60 iterations, because at a singular solution the rate $\rho/(\sigma_{\min}^2 + \rho)$ of §4.6 tends to 1.

(k) An unreachable target. Take $\mathbf{y}^* = (0.8, 0.1)$. Its radius exceeds the arm's reach:

$$\|\mathbf{y}^*\| = 0.8062 > 0.7.$$

The globally smallest residual follows from the reachable annulus. It is $0.106226\ \mathrm{m}$, at the closest point

$$\mathbf{p}_c = (0.6946, 0.0868), \qquad \mathbf{q} = (7.125^\circ, 0^\circ), \qquad \nabla f = 0.$$

Plain Gauss–Newton nearly straightens the arm, then requests a huge step. Its first iterate is $(-26.6^\circ, 66.3^\circ)$ with residual 0.229. Its second is $(7.6^\circ, 0.8^\circ)$ with residual 0.107. The third step has norm $49.8\ \mathrm{rad}$ and moves to $(1131^\circ, -2623^\circ)$. The residual subsequently ranges between 0.1 and 1.3. If left to run, it reports `diverged` at iteration 364 with a $3540\ \mathrm{rad}$ step.

Armijo and damping affect different parts of this process. The following runs start from the same configuration:

Method	Outcome	$\mathbf{q}$ (degrees)	Residual (m)	Excess above the minimum	Tip distance to $\mathbf{p}_c$	$\|\nabla f\|$
Gauss–Newton with Armijo, floor 10^{-6}	`stalled` at iteration 7	$(7.93, 0.02)$	0.106763	$0.54\ \mathrm{mm}$	$9.96\ \mathrm{mm}$	8.7×10^{-3}
Damped, $\rho = 0.01$, full steps	`stationary` at iteration 29	$(7.125, 0.000)$	0.106226	$10^{-17}\ \mathrm{m}$	$0.00\ \mathrm{mm}$	5.3×10^{-10}

The Armijo run stops about 1 cm from the closest point, with a nonzero gradient. Its shoulder angle differs by 0.8°. Its residual is only about 0.5 mm above the minimum, so residual alone hides the remaining position error. Along this boundary, a 1° shoulder change moves the tip 12.2 mm but raises the residual only 0.81 mm. Damping changes the direction and reaches the closest point in this run.

Optional numerical detail on the line-search floor. At the stall, the Gauss–Newton direction is 89.96° from downhill, within 0.05° of perpendicular, and about 1.7×10^3 rad long. The trial at $\alpha = 2^{-20}$ raises the cost by 8.6×10^{-9} while Armijo requires a change at most -1.1×10^{-10}. It is the first trial at or below the 10^{-6} floor, so the solver rejects it after six accepted steps and reports iteration 7. The next halving, 2^{-21}, would pass. It lowers the cost by 5.6×10^{-10} and moves the joints by 8×10^{-4} rad. The stopping event therefore belongs to the floor, while the poor progress belongs to the direction.

Floors of 10^{-8} and 10^{-12} stop at iterations 10 and 12. Both endpoints remain 9.96 mm from the closest point, with gradient norm 8.7×10^{-3}. With a floor of 10^{-15}, the normal matrix becomes numerically singular. Forming $\mathbf{J}^\top\mathbf{J}$ squares the conditioning of $\mathbf{J}$; QR or singular-value decomposition of $\mathbf{J}$, as used by `np.linalg.lstsq`, avoids forming that matrix. A failed Armijo trial must be rejected. Taking it anyway or adding a gradient fallback would change the algorithm used here.

Optional numerical detail on damping. Enough damping is required for this target. With $\rho = 0.001$, full steps raise the residual 50 times in 100 iterations and end far from the boundary. With $\rho = 0.01$, the relative gradient test stops the run at iteration 29 with gradient norm 5.3×10^{-10} and the tip within 10^{-5} mm of the closest point. Its step falls below 10^{-6} rad from iteration 24. With $\rho = 0.1$, the stationary outcome arrives at iteration 99 with gradient norm 7.7×10^{-10}; its step is below 10^{-6} rad from 71. Adaptive damping gives steps below 10^{-6} from iteration 20 and the stationary outcome at 24.

The small-step observations are not stops because that check is off. The separate 100-iteration runs also turn the stationarity stop off. For $\rho = 0.01$, the gradient norm after 100 iterations is 7×10^{-19}. Section (k) of the script prints all these variants.

(l) The set of fixed points. From $\mathbf{q}^{(0)}$, $\rho = 0$ and $\rho = 0.5$ end at the same $\mathbf{q}$ to 4×10^{-6} rad. On a 35×35 grid of starts over $[-180^\circ, 180^\circ]^2$ the two damping values reach the same solution from 814 starts, different solutions from 384, and one of them fails from 27 ($\rho = 0$ fails from 27 of them, $\rho = 0.5$ from none): $\mathbf{D}$ changes the path and sometimes the valley, never the set of fixed points, and it can rescue a run (§4.9).

(m) What the cost does not know. Both solutions bend the elbow to $|q_2| = 80.4^\circ$. With an elbow stop at 60°, clipping q_2 to $[-60^\circ, 60^\circ]$ after each Gauss-Newton step, stalls at $(-16.40^\circ, 60^\circ)$ with the tip at $(0.601, 0.094)$ and residual 0.1464 m, which is 4.4 times the cost of the best allowed configuration (its residual is 0.0698 m , Deck 05 §5 (b)). A limit is a constraint, and Deck 05 handles it as one.

(n) The Woodbury form. At $\mathbf{q}^{(0)}$ with $\mathbf{K} = \mathbf{I}$, $\mathbf{D} = 0.05\mathbf{I}$, the two forms (18) and (19) of $\mathbf{J}^\sharp$ agree to 1.3×10^{-15}.

(o) The same step on a real arm. The Panda example uses Drake's forward kinematics and position Jacobian, with three task coordinates and seven joints. At its regular ready pose, $\mathbf{J}^\top\mathbf{J}$ is a 7×7 matrix of rank 3. The undamped normal equations therefore have infinitely many solutions. They remain consistent because $\mathbf{J}^\top\mathbf{e}$ lies in the range of $\mathbf{J}^\top\mathbf{J}$. Their missing directions are exactly the Jacobian's null space:

$$\mathbf{J}^\top\mathbf{J}\boldsymbol{\delta} = 0 \quad \Longleftrightarrow \quad \|\mathbf{J}\boldsymbol{\delta}\|^2 = 0.$$

Damping makes the step unique. The pseudoinverse instead selects the minimum-norm least-squares step. The supplied script, `drake/02-gauss-newton-ik/gn_ik_panda.py`, compares three targets.

For the first target, move 10 cm sideways and 5 cm down from the ready pose. With $\rho = 10^{-3}$, the residuals are

$$1.1 \times 10^{-1}, \quad 1.7 \times 10^{-2}, \quad 3.7 \times 10^{-4}, \quad 3.3 \times 10^{-6}, \quad 5.0 \times 10^{-8}.$$

The run solves the target in four steps. Rapid early convergence is followed by the damping's linear rate. The late residual ratio is 0.015, close to the predicted 0.016 with smallest singular value 0.25 (§4.6).

For the second target, start from a horizontal stretched pose with the elbow at its straight limit. The smallest singular value is 0.095. Request a further 5 cm in the weakest tip-motion direction. With $\rho = 10^{-3}$, the run needs 18 steps to a residual below 10^{-6}. At the solution, the smallest singular value is 0.026 and the rate is

$$\frac{\rho}{\sigma_{\min}^2 + \rho} = 0.596.$$

The solution is regular. Reducing damping to 10^{-8}, or using the pseudoinverse, solves the same target in five steps. The 18 steps reflect damping along a weak direction; they do not reproduce the two-link arm's singular boundary regime.

For the third target, request 30 cm in that direction. The plain steps have norms between 0.9 and 4.2 rad and never settle. The run ends with joint 2 at 4.06 rad, beyond its model limit of 1.76 rad. The robot description is a Unified Robot Description Format (URDF) file. The squared position cost does not enforce its joint bounds.

Adaptive damping finds a numerical reference with residual $0.24621\ \mathrm{m}$ and gradient norm 10^{-9}. This reference is a stationary point rather than a certificate of the globally smallest task error. With Armijo on the Gauss–Newton direction, no trial reaches the rejection floor in 40 iterations. At the iteration limit, the residual is $0.24621\ \mathrm{m}$, only $9 \times 10^{-7}\ \mathrm{m}$ above the reference. The tip is 0.6 mm from the reference tip, the gradient norm is 7.5×10^{-4} and the step is still 0.003 rad. This run progresses

slowly; it is neither stalled nor stationary to the stated tolerance. Fixed damping $\rho = 0.01$ also fails to settle, with residual 0.285 after 2000 steps, while the adaptive rule does.

Joint limits are not enforced in any of these Panda runs. The all-zero Panda pose still permits three independent translational motions, so position-only IK does not stall there as it does for the stretched two-link example. The Drake guide records the distinction.

(p) A reachable target where the standard local steps are zero. At the stretched arm $\mathbf{q} = (0, 0)$ the tip is $(0.7, 0)$. Take the reachable target $\mathbf{y}^* = (0.5, 0)$ ($\|\mathbf{y}^*\| = 0.5$, inside the annulus $[0.1, 0.7]$): $\mathbf{r} = \varphi - \mathbf{y}^* = (0.2, 0)$, $\mathbf{J} = \begin{bmatrix} 0 & 0 \\ 0.7 & 0.3 \end{bmatrix}$, and $\nabla f = \mathbf{J}^\top \mathbf{r} = (0, 0)$. The exact Hessian there has eigenvalues -0.0293 and 0.4093, so the point is a saddle of f, not a minimum. The standard selected steps are zero: the minimum-norm Gauss-Newton step (the pseudo-inverse, since $\mathbf{J}^\top \mathbf{J}$ is singular), the damped steps at $\rho = 10^{-3}, 10^{-2}, 10^{-1}$, and the gradient step. `run_gn` from there reports "stationary" with a residual of 0.200 m. The target has two solutions, $(-36.87^\circ, 90^\circ)$ and $(36.87^\circ, -90^\circ)$. Every start near the singular line is nearly singular, so the first plain step is large: from a bend of 10° plain Gauss-Newton reports "solved" in 9 steps at the second solution, but only after a first step of 7.26 rad (from -10° the mirror), and from a bend of 1° the first step is 72.7 rad (the lesson of (k)) before it lands on a solution in 7 steps. The protected runs from the 10° bend show how step control changes the outcome: with Armijo, "solved" in 4 steps at $(-36.87^\circ, 90^\circ)$ with a first step of 1.81 rad; with $\rho = 0.01$, "solved" in 8 steps with every step below 0.84 rad (all printed by (p)). So a zero step is neither success nor proof that no $\mathbf{q}$ reaches $\mathbf{y}^*$; unreachability is shown by geometry. Figure `fig/stationary_straight.png` draws the arm, the target on its line, the collapsed velocity ellipse, and the elbow-bent solution. This example separates local stationarity from target reachability.

6. Code

The supplied scripts reproduce the worked examples. Run them from the repository root with the course Python environment described in `SETUP.md`:

```
.venv/bin/python lectures/03-D1-optimization-1/code/descent.py
.venv/bin/python lectures/03-D1-optimization-
        1/code/gauss_newton_ik.py
.venv/bin/python drake/02-gauss-newton-ik/gn_ik_panda.py
```

`descent.py` prints the quadratic, Armijo and Newton calculations of §5(a–d). `gauss_newton_ik.py` prints the two-link calculations of §5(e–n,p). Its `phi` and `jac` functions implement forward kinematics and its Jacobian. `run_gn` returns the iterates

and an outcome; `report()` prints the endpoint quantities. With `linesearch=True`, the solver keeps the Gauss–Newton direction and uses Armijo to choose its length. A failed trial at the floor ends that run as `stalled`; smaller floors are compared in §5(k). The Panda script uses Drake for forward kinematics and the Jacobian, then applies the same update. Add `--meshcat` to animate its iterates with the target drawn as a sphere.

The Gauss–Newton browser module lets you compare plain Gauss–Newton, fixed damping and gradient descent. Predict the next tip position, take one step and compare the actual residual with that prediction. Its default target is $(0.75, 0.05)$ m, which differs from §5(k)'s target. The module's Armijo floor is 10^{-8}; the script's default is 10^{-6}. Gradient descent always uses Armijo, while the other directions have an optional line search. The module notes describe the controls and outcomes.

The optimization explorer compares directions on the elongated quadratic. Start with gradient descent and line search off. Change the step size and observe contraction, oscillation or divergence. Then enable Armijo and compare the accepted steps. Newton reaches the quadratic's minimizer in one step. The app writes $\mathbf{x}$ for this reading's $\mathbf{z}$ and λ for damping ρ. Its Diagnostics row reports the previous accepted step size; the next search starts from a grown trial value and may shorten it. Its preview shows that next accepted step.

The Drake guide gives the Panda command and controls. `code/make_figures.py` regenerates the static examples from the same functions. The figures and numerical tables supply the explanation when an interactive scene is unavailable.

7. Common failure modes

- **Treating the step size as a hyperparameter to tune.** It is a per-step decision; (3) makes it. A fixed α that works at one $\mathbf{q}$ diverges at another because $\mathbf{H}$ changes.
- **Scaling.** Changing meters to millimeters in one coordinate can worsen $\operatorname{cond}(\mathbf{H})$ by a factor of 10^6; gradient descent then crawls, an exact Newton step is invariant under a consistent affine change of variables, while floating-point linear solves remain sensitive to conditioning. Rescale variables, or use a Newton-type method. The same fact bites the step size: α has units (§3.4), and a cap of $\alpha \leq 1$ cost the arm 312 iterations instead of 46 (§5 (g)).
- **Trusting the linear model far from the iterate.** The 1.59 rad first step of §5 (f) (74.1° of it at the elbow) and the 49.8 rad step of §5 (k) are the same mistake at different severities. Armijo accepts the first (the cost falls by two thirds) and rejects the second; damping shortens both.
- **Confusing the step direction with the gradient.** The Gauss-Newton step is $-(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top\mathbf{r}$, not $-\mathbf{J}^\top\mathbf{r}$ (that is gradient descent) and not $-\mathbf{J}$ (a shape error: an $m \times n$ matrix where an n-vector is needed).

- **Reading a stationary point as a solution, or as a proof.** At an unreachable target the stretched arm has $\mathbf{J}^\top(\mathbf{y}^* - \varphi) = 0$ with residual $0.106\,\mathrm{m}$, the distance to the reachable set (§5 (k)); at a reachable one the same arm can be stationary with residual $0.2\,\mathrm{m}$ (§5 (p)). Report both $\|\varphi(\mathbf{q}) - \mathbf{y}^*\|$ and $\|\mathbf{J}^\top(\varphi(\mathbf{q}) - \mathbf{y}^*)\|$ and the outcome word (§4.7), never just "converged", and never read "stationary" as "unreachable".
- **Reading a floor as a theorem.** "Stalled" is the implementation's stopping rule: the trial at its floor failed. A smaller trial can pass (§5 (k)), and the slow progress belongs to the direction, not to the floor.
- **Forming $\mathbf{J}^\top\mathbf{J}$.** The normal equations square the conditioning of $\mathbf{J}$; the code forms them because the derivation does, and §5 (k) shows the matrix turning numerically singular at a floor of 10^{-15}. A QR or SVD of $\mathbf{J}$ (`np.linalg.lstsq`) is the numerically preferable solve.
- **Expecting a line search to do damping's job.** Armijo on the Gauss-Newton direction keeps the step finite and can stop where the gradient is not zero (§5 (k)); damping changes the direction. The $\rho = 0.01$ run in that example reaches the stationary point. They act on different decisions.
- **Dropping the damping on a redundant arm.** $\mathbf{J}^\top\mathbf{J}$ is singular for every $\mathbf{q}$ when $n > m$, so the normal equations have infinitely many solutions; use $\mathbf{D} \succ 0$ (then (18) or (19)) or the minimum-norm $\mathbf{J}^+$.
- **Expecting one answer.** The two-link arm has two; a 7-joint arm has a continuum. The start and $\mathbf{D}$ choose.

8. Where this sits in the stack

These methods implement the stack's Goal box. Perception supplies a target, the robot model supplies forward kinematics and its Jacobian, and the local solver proposes joint configurations that reduce task error.

Optimization II extends the formulation to a trajectory with costs on velocities and accelerations, together with joint-limit and clearance constraints. The residual structure still supports Gauss–Newton models, and damping keeps the inner step well posed.

A learned sampler can instead propose joint configurations or starting guesses. An imitation policy can output joint or task-space commands directly. The action format does not establish that the network has learned an IK solver. Its commands still need the appropriate geometry, frame conventions and feasibility checks. Training coverage also determines whether it works at an unfamiliar target or robot configuration.

9. Further reading (optional, never load-bearing)

- Jorge Nocedal and Stephen Wright, *Numerical Optimization*, chapters 2–3 (line search, Wolfe conditions) and 10 (least squares, Gauss-Newton, Levenberg–Marquardt).
- Marc Toussaint, *A tutorial on Newton methods for constrained trajectory optimization* (2017): the KOMO view of the same Gauss-Newton step, used from 4.2 on.
- Russ Tedrake, *Robotic Manipulation*, chapter 3, "Differential inverse kinematics as an optimization": the same problem with joint-limit constraints, which is week 4's opening.
- `source-2026/opt-lectures/01-03*.md` (frozen): last year's fuller treatment, including Wolfe bracket-and-zoom, steepest descent in a general norm, and the affine-invariance argument.

10. What remains beyond this lesson

A local model supplies a proposed step. Line search checks actual cost reduction, and damping penalizes the proposed step. Neither method establishes that the target is reachable or guarantees convergence from every starting guess.

The worked inverse-kinematics costs use position error. A full-pose objective also needs a chosen orientation error and its derivative, with its expression frame and sign defined. Joint bounds and collision requirements are constraints rather than facts known by the squared position error. Optimization II develops those constraints.